How do Osimertinib treatment affect different type of genes in cancer cells

Bioskills — Final Assignment

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Adapted from: Oren et al., Nature (2021)

PC9 lung cancer cells were treated with Osimertinib and profiled at four timepoints by scRNA-seq

The question

How does gene expression change over time,and do cycling and non-cycling cells diverge at Day 14?

The experiment

Cell line: PC9 (non-small cell lung cancer)

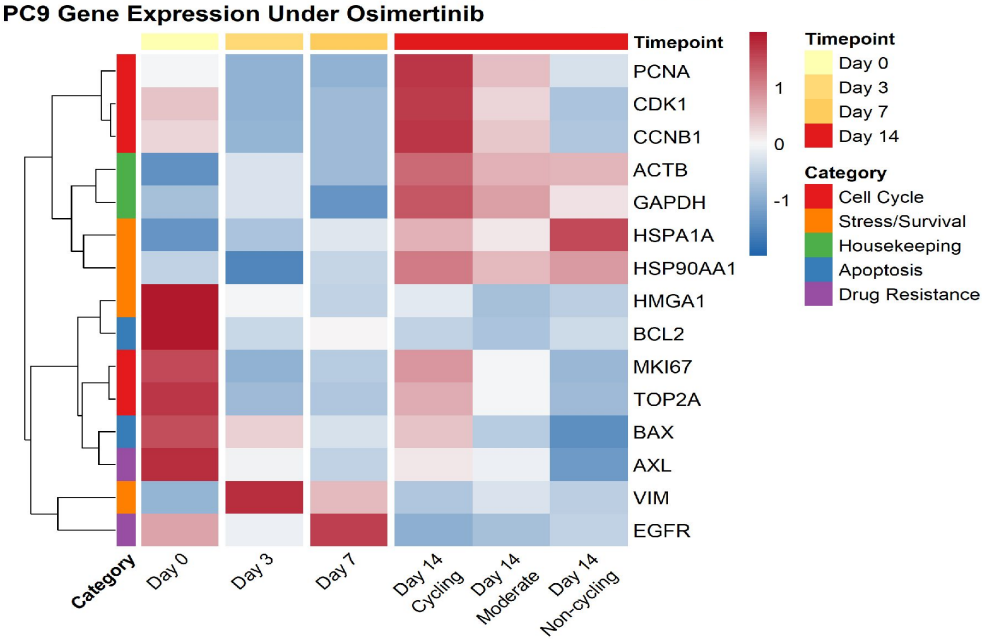
Drug: Osimertinib (EGFR inhibitor)

Group checked: Day 0, 3, 7, and 14 (cycling, moderately cycling and non cycling separated by FACS)

scRNA-seq: ~56,000 individual cells profiled

Genes: we choose 15 relevant genes to check the effect of the drugs on them

The drug reshapes gene expression within 3 days, and make significant change between cycling and non cycling cells in 14 days



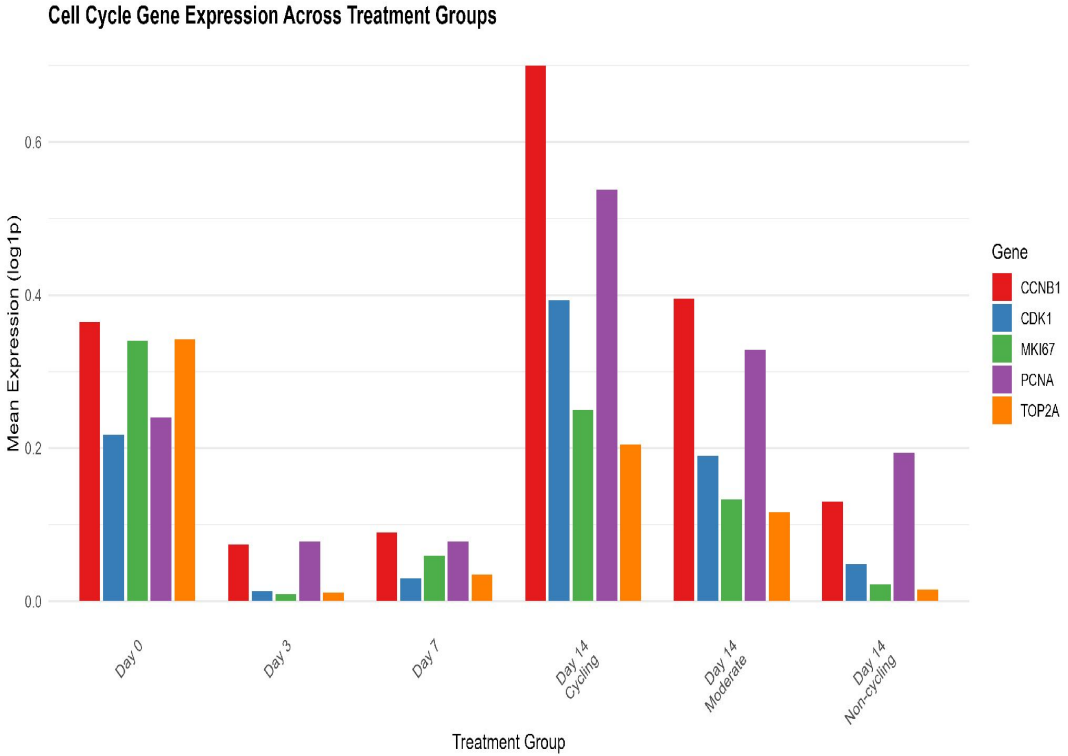
Insight

- **Rapid Initial Action:** The drug acts fast, showing a sharper decline from Day 0 to Day 3 than from Day 3 to Day 7.
- **Transition Stress:** VIM expression peaks at Day 3 and then fades, indicating a temporary acute stress response rather than a permanent shift.
- **Day 14 Divergence:** Populating cells split into two distinct trajectories:
  - **Cycling cells:** Reactivate cell-cycle-associated genes.
  - **Non-cycling cells:** Maintain an arrested state with elevated stress markers (HSPA1A).

Change made to the data

- **Gene Selection:** Reduced the dataset from tens of thousands of genes down to 15 relevant, role-categorized genes for computational efficiency (the full dataset failed to plot after an hour of running).
- **Data Aggregation:** Averaged the scRNA-seq profiles across replicates for each day to generate a cleaner visualization, as the individual results were highly consistent.

Zoom in on the cell cycle associated genes through the 14 Days, and the difference between different groups at 14 Days



Key findings

- **Day 3 :** Proliferation is almost completely suppressed by the drug.
- **Day 14 (Resistance):** Cycling cells exceed Day 0 levels (CCNB1, PCNA) due to natural selection of a resistant subpopulation.
- **Arrest Spectrum:** Not binary; exhibits a clear gradient Cycling > Moderate > Non-cycling.